

NAME: _____ MATRIC NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 1 SEMESTER 1 2021-2022

MS1016 – THERMODYNAMICS OF MATERIAL

20 September 2021

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
2. Answer **ALL** questions.
3. This is an open book test.
4. All questions have **ONLY ONE** correct answer for the MCQ questions.

PART I – SHORT QUESTIONS

- (1) Explain the term ‘Isentropic’ condition. Name a process that is isentropic for a system under a specific condition. (20 marks)

Isentropic process is a process where the entropy is constant. This means the total entropy change (system and surrounding).

$$dS = 0$$

Isentropic process is an idealized process in which the process is totally reversible. By Carnot’s theorem,

$$dS = \frac{\delta Q}{T}$$

Based on the above equation, for a reversible process, isentropic process must imply $\delta Q = 0$. This mean that no heat is absorbed or released. This condition is also called the adiabatic process. Therefore, an isentropic process is also an adiabatic process when it is reversible. If the process is irreversible, then

$$dS = \frac{\delta Q}{T} > 0$$

This means $\delta Q > 0$ implying that the process is not isentropic

2. Given $dH = TdS + VdP$. Show that

$$\left(\frac{\partial H}{\partial P}\right)_T = V(1 - \alpha T)$$

where α is the isobaric coefficient of thermal expansion. V and T are the volume and temperature respectively. H is the enthalpy, P , the pressure and S , the entropy. Given: $dG = VdP - SdT$.

(20 marks)

Given

$$dH = TdS + VdP$$

$$\left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V \left(\frac{\partial P}{\partial P}\right)_T$$

$$\left(\frac{\partial H}{\partial P}\right)_T = T \left(\frac{\partial S}{\partial P}\right)_T + V$$

From Maxwell' relationship

$$\left(\frac{\partial H}{\partial P}\right)_T = -T \left(\frac{\partial V}{\partial T}\right)_P + V$$

$$\left(\frac{\partial H}{\partial P}\right)_T = -T\alpha V + V$$

$$\left(\frac{\partial S}{\partial P}\right)_T = V(1 - \alpha T)$$

3. Why do solid melts when heated above the melting point?

(20 marks)

The Gibbs free energy, G is given as

$$G = H - TS$$

Where H is the enthalpy and S the entropy. T is the temperature.

It can be seen from the Gibbs free energy equation that the state with the highest stability will be that with the best compromise between low enthalpy and high entropy. At low temperatures, solid phases are most stable since they have strongest atomic bonding and therefore lowest internal energy (enthalpy). At high temperatures, the term $-TS$ dominates and phases with more freedom of atom movement, gases and liquids become most stable.

PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(50 marks)

- (a) For an ideal gas, it is given that $\left(\frac{\partial H}{\partial P}\right)_T = 0$. This implies that

- A Enthalpy is not a function P
- B Enthalpy is not a function of T
- C Enthalpy is a function of P and T
- D Enthalpy is not a function of P and T
- E Enthalpy is a function of P

ANS: A

- (b) In the _____ compression and expansion of an ideal gas, the temperature decreases due to decreasing _____ .

- A Isobaric, pressure
- B Adiabatic, pressure
- C Isothermal, enthalpy
- D Adiabatic, entropy
- E Adiabatic, enthalpy

ANS: B

- (c) 2 rectangular pieces of brick with different mass and temperature come into contact at one face. If the 2 bricks are completely isolated from the surrounding, which of the following is likely to be true at equilibrium?

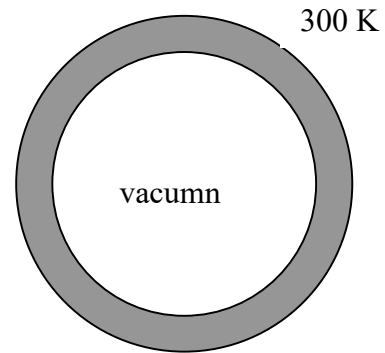
- A They have different temperature
- B Their combined entropy is maximum
- C Their respective internal energy are constant
- D The combined enthalpy is maximum
- E Work done is PdV

ANS: B

- (d) A perfectly insulated, evacuated, rigid sphere is surrounded by air at 300 K and 1 atm. Suddenly a hole is punched in the sphere. What will happen?

- A the temperature inside the sphere will be 300 K
- B the pressure inside will decrease
- C the temperature inside will decrease
- D the temperature inside will increase
- E none of the above

ANS: D



- (e) “in all energy exchanges, if no energy enters or leaves the system, the potential energy of the state will always be less than that of the initial state.” The statement is synonymous to:

- A First Law of Thermodynamics
- B Second Law of Thermodynamics
- C Third Law of Thermodynamics
- D Zeroth Law of Thermodynamics
- E Gibbs Criteria for Equilibrium

ANS: B

- (f) “energy can be changed from one form to another, but it cannot be created or destroyed. The total amount of energy and matter in the Universe remains constant, merely changing from one form to another”. The statement is synonymous to:

- A First Law of Thermodynamics
- B Second Law of Thermodynamics
- C Third Law of Thermodynamics
- D Zeroth Law of Thermodynamics
- E Fourth Law of Thermodynamics

ANS: A

(g) What is the heat of formation of a compound?

- A energy evolved or absorbed during the formation of the compound
- B internal energy of the compound
- C entropy change in the formation of the compound
- D enthalpy change in the formation of the compound under changing pressure
- E mass change in the formation of a compound

ANS: A

(h) Which of the following statements is most likely to be TRUE?

- A The entropy of a system cannot be decreased
- B The entropy of a system can only be increased by decreasing the entropy of the surrounding
- C The entropy of a system can be decreased by interacting with the surrounding
- D The total entropy of the system and surrounding is constant
- E Spontaneous reaction is associated with a decrease in entropy

ANS: C

(i) When the pressure on a solid with positive volumetric coefficient of thermal expansion, is increased abruptly,

- A the volume decreases
- B the volume increases
- C the temperature of the solid remains constant
- D the temperature of the solid increases
- E the temperature of the solid decreases

ANS: D

(j) The rate of decrease in Gibbs free energy with temperature at constant P equals to the _____ of the system:

- A enthalpy
- B internal energy
- C pressure
- D volume
- E entropy

ANS: E